

CHEMICALS IN COSMETICS

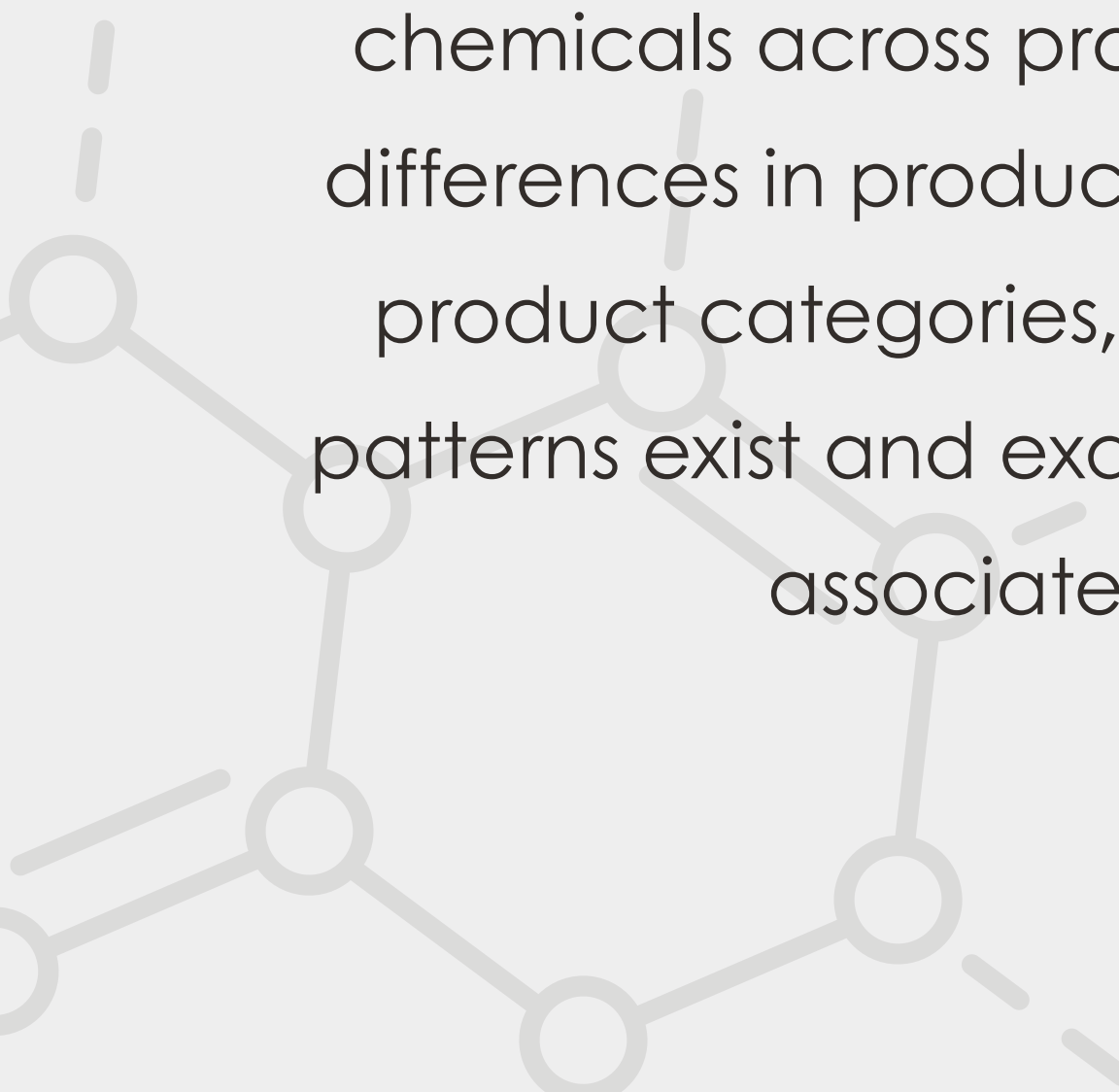
Chemical in Cosmetics in the state of California

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Problem Statement

The California cosmetic chemical disclosure dataset contains 112,870 records reported between 2009 and early 2020. The data shows differences in the number of reported chemicals across product categories, changes in disclosure volume over time, and differences in product discontinuation. This project analyzes the reported chemicals, product categories, reporting trends and discontinuation information to identify if patterns exist and examine whether the number and type of reported chemicals are associated with product characteristics and discontinuation.



Key matrices

13

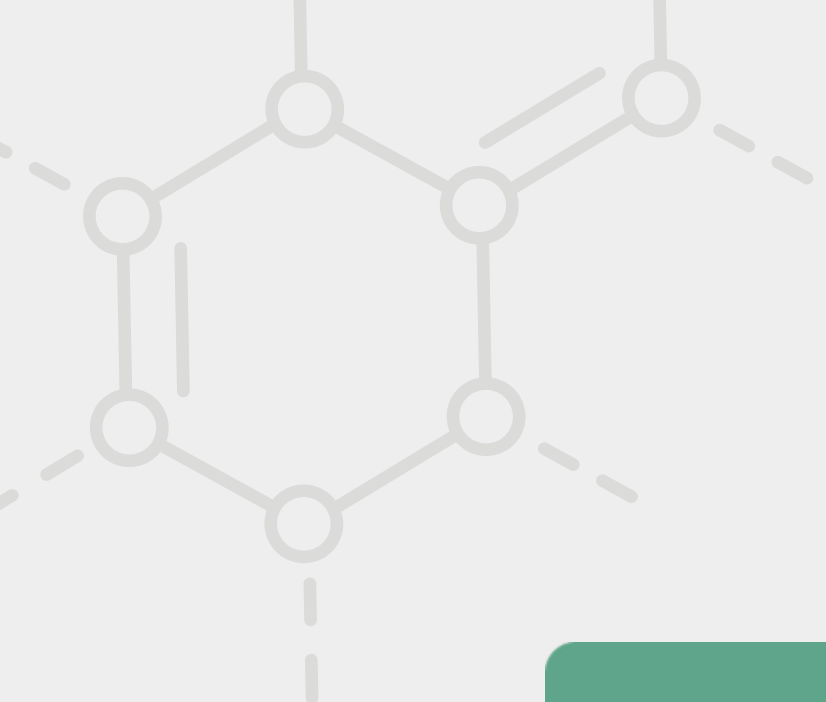
Product categories

2009 – Feb 2020

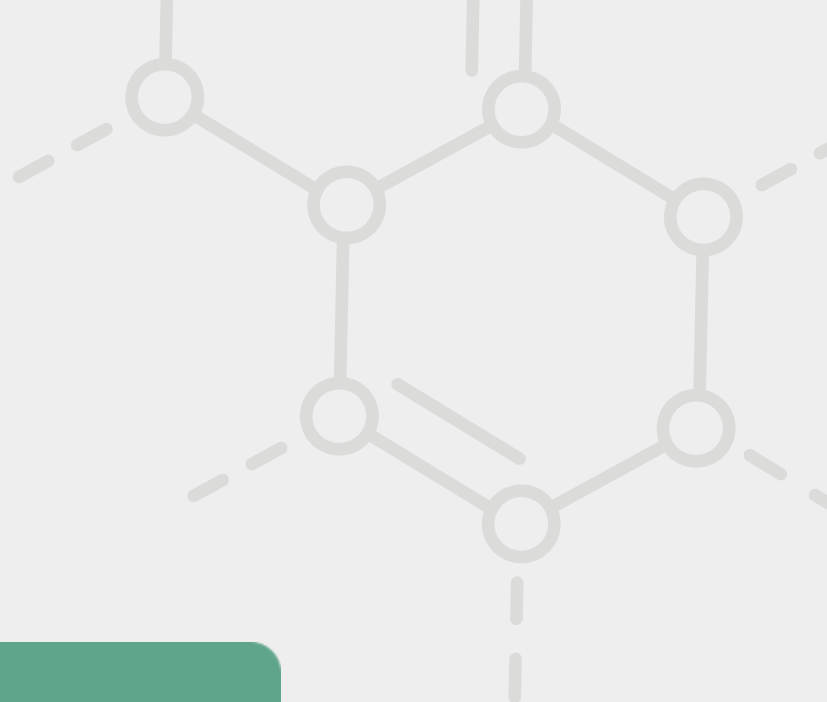
period covered

California, USA

Location



Key Terms



Primary category: the product group, such as Makeup, Nail or Skincare

Discontinued: the product has a discontinuation date on record

Discontinuation rate: the share of products that were discontinued

Objectives

Which product categories have the most chemicals and does the pattern hold when accounting for the number of products in each category?

Does the number of chemicals per product vary by category?

Which chemicals are most commonly reported overall, and how does the most commonly vary across products?

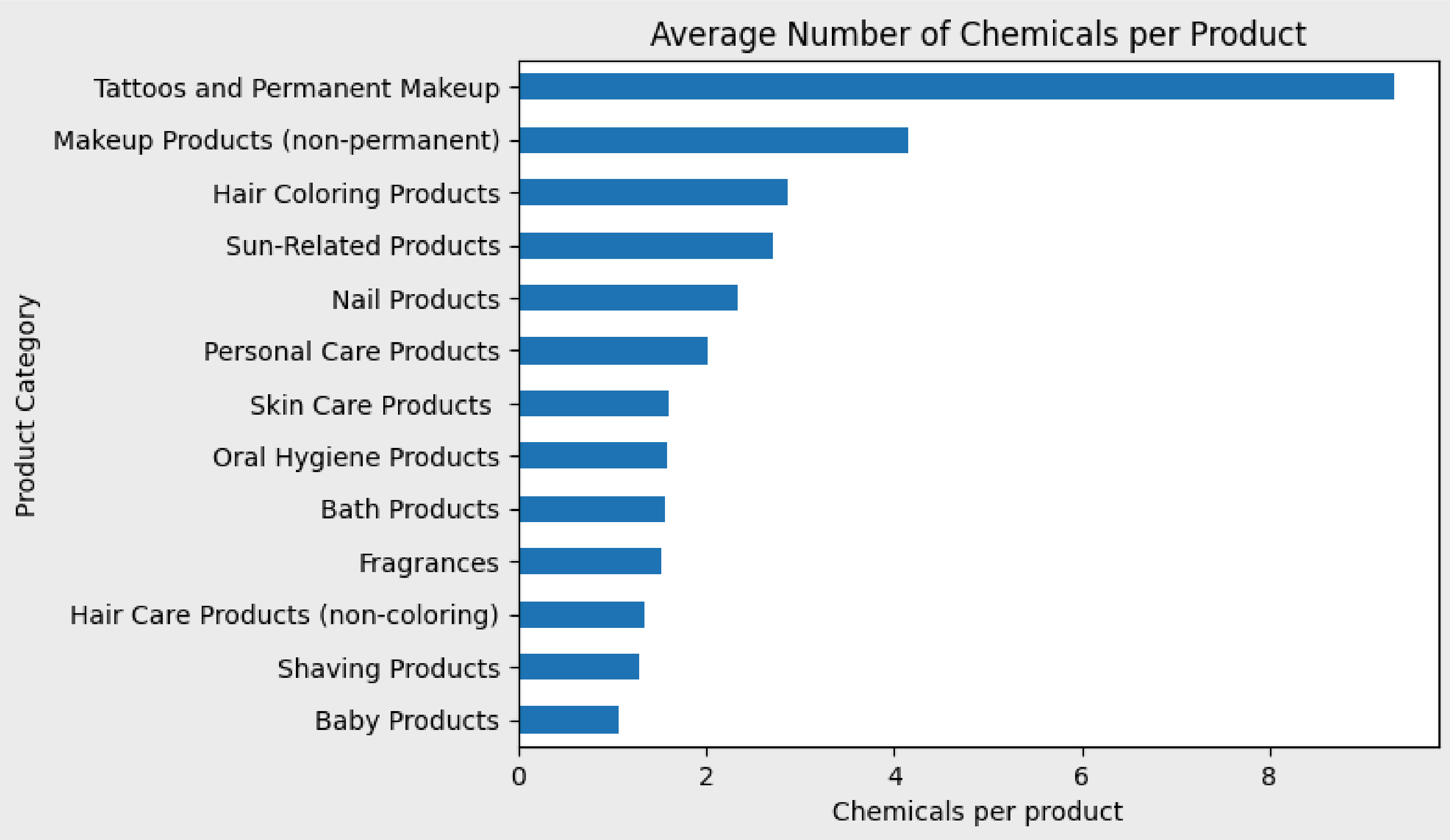
How does the rate of product discontinuation vary across categories?

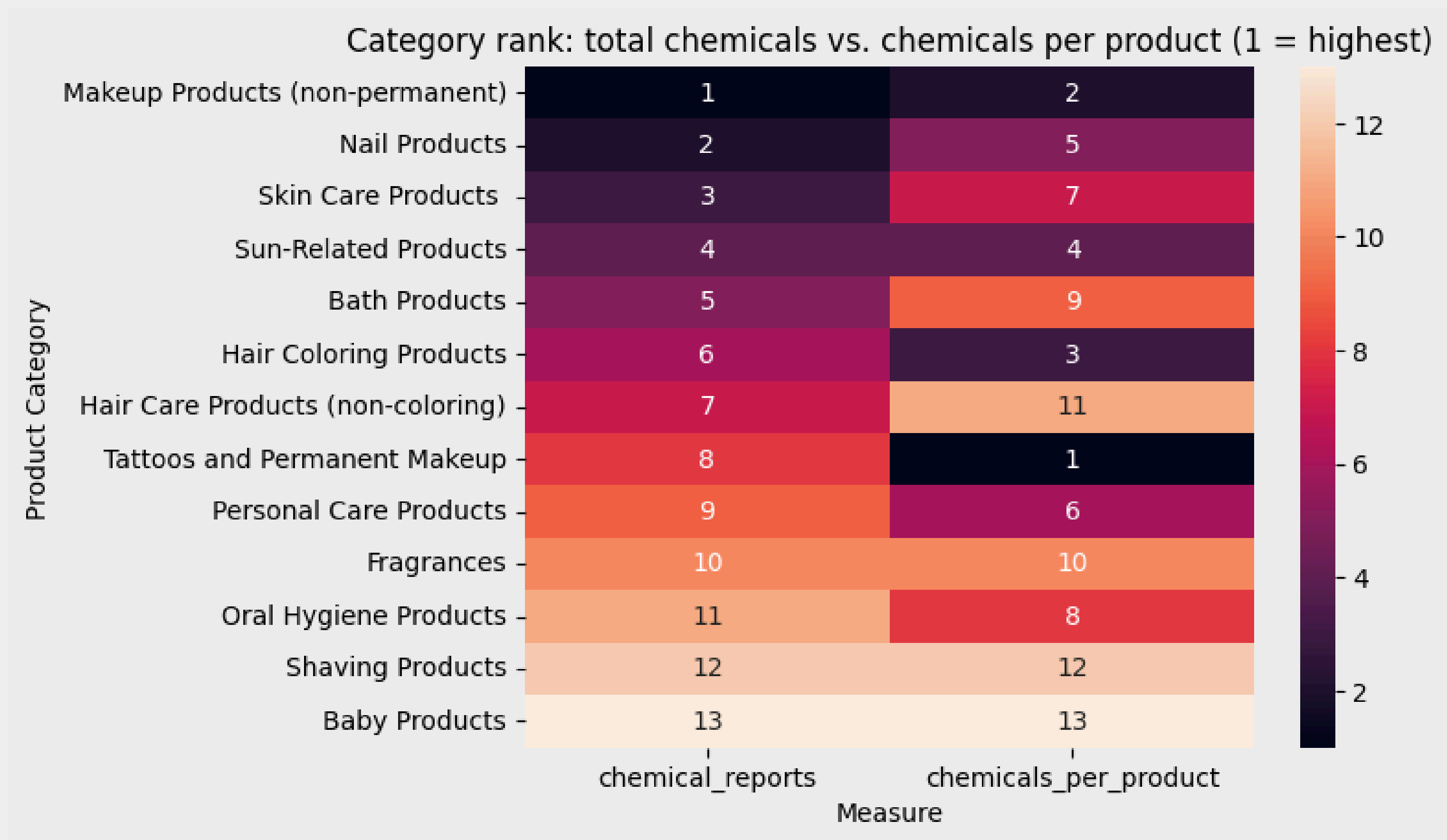
Has the volume of disclosure changed over time?

Are specific chemicals more commonly associated with discontinued products?

Is there a relationship between the number of chemicals in a product and the discontinuation?

Average Chemicals per product

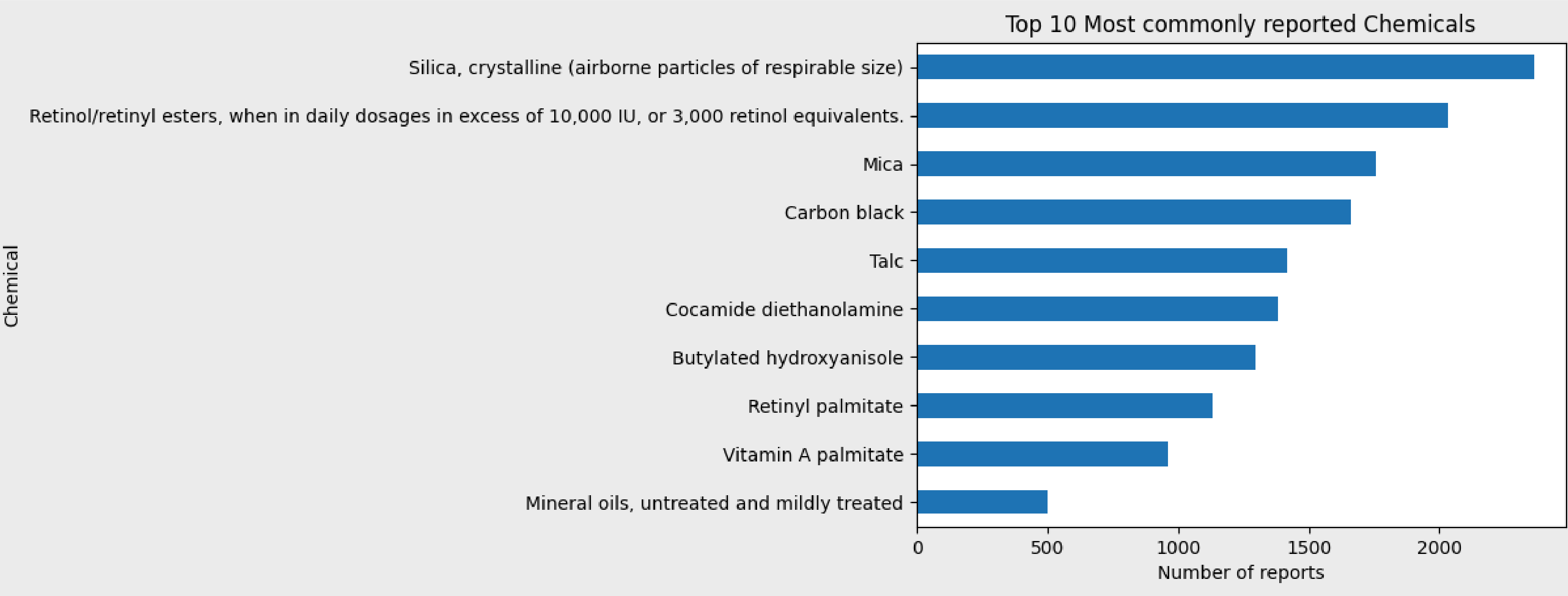




Findings

- Makeup has the most chemical reports: 68.7K, or 65% of all reports.
- The ranking does not hold once products are counted: totals and per-product rates give different ranks.

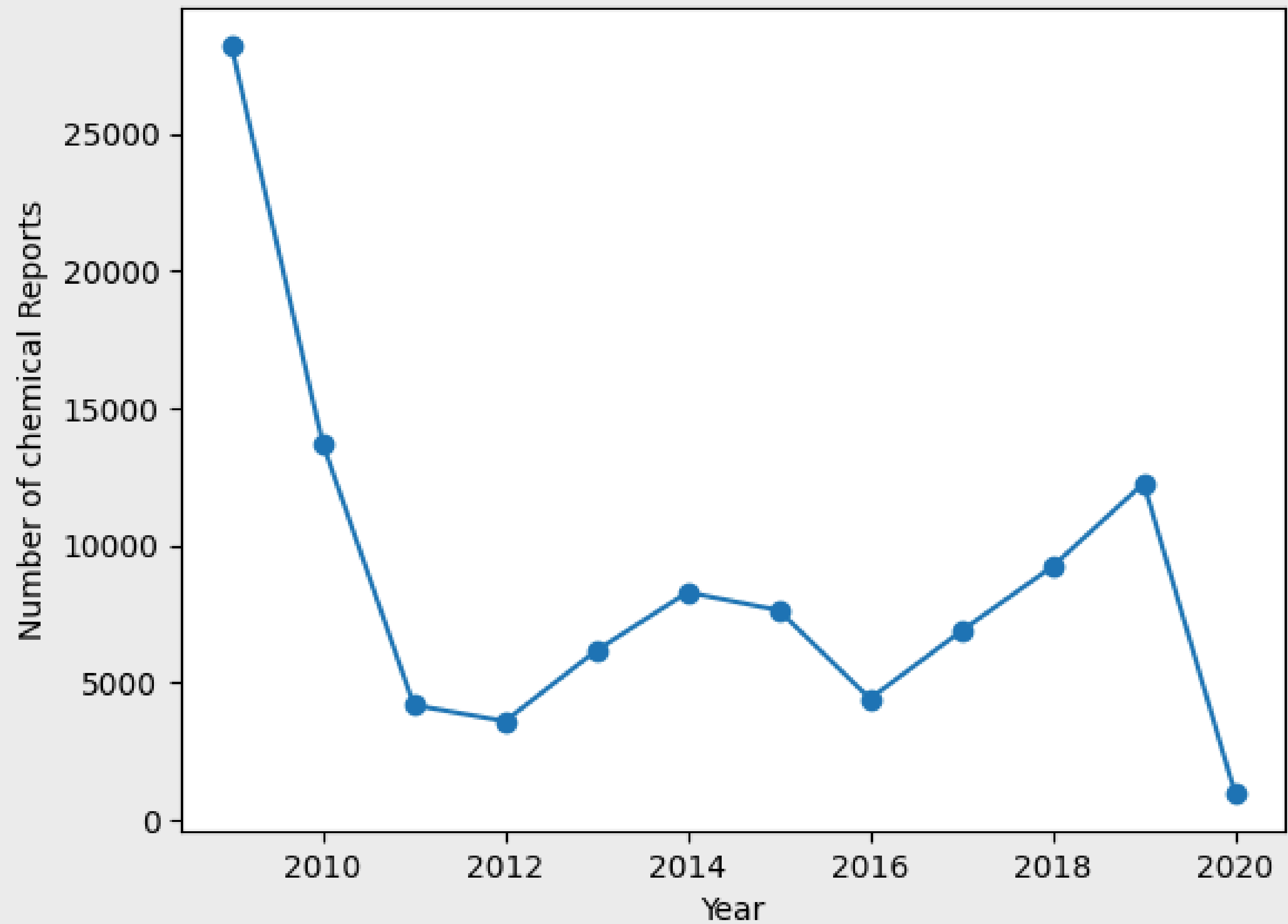
Most Reported Chemicals



Findings

- Titanium dioxide makes up 82% of all reports and is the top chemical in all 13 categories.
- Excluding it top chemicals differ by category: mica (Makeup), crystalline silica (Nails), cocamide DEA (Bath, Hair Care), retinol (Skin Care), carbon black (Tattoos).
- 123 distinct chemicals appear in total.

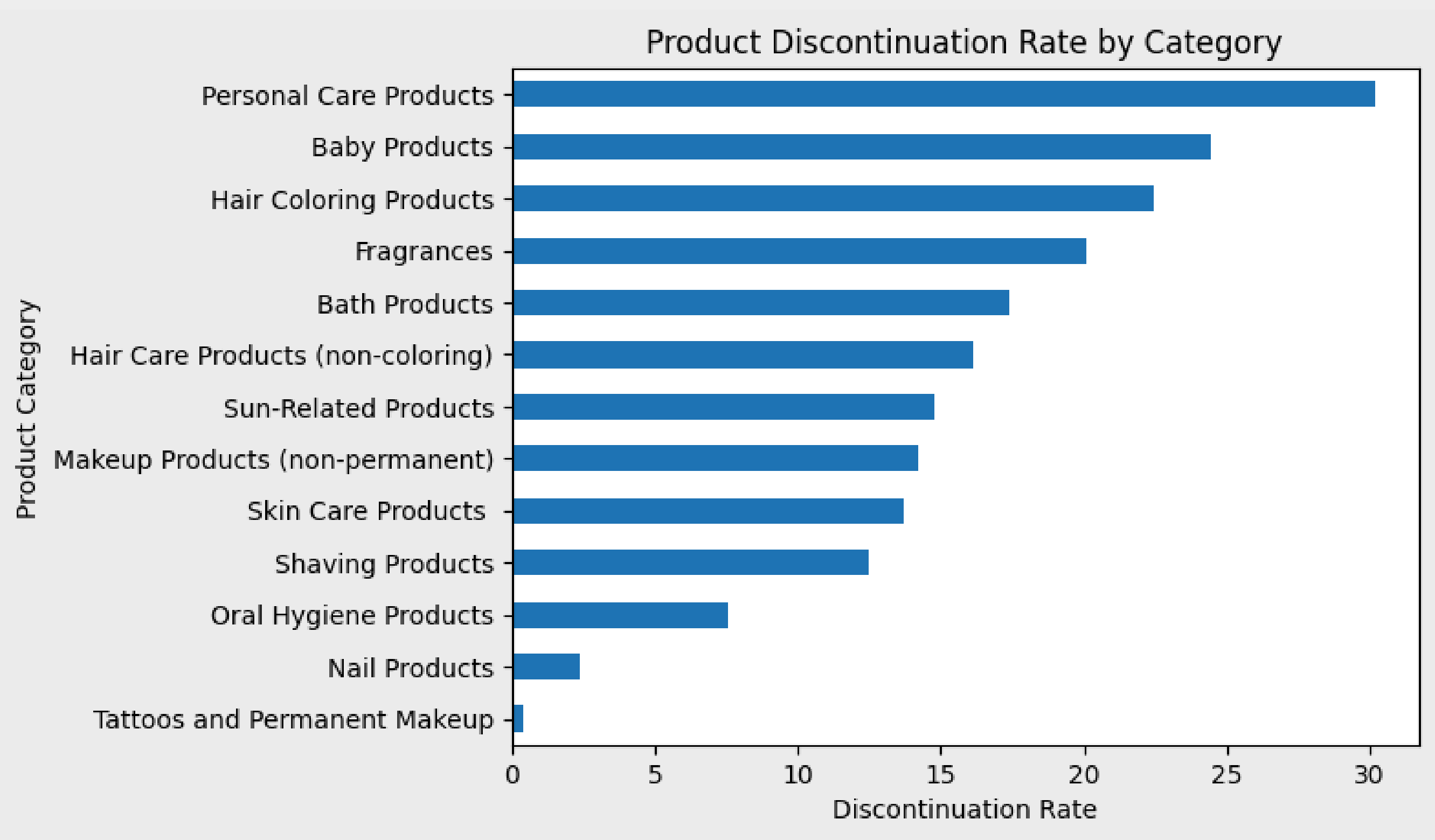
Chemical Disclosure Volume Over Time



Findings

- Reports fell 87% from 28.2K in 2009 to 3.6K in 2012.
- They have since recovered to 12.3K in 2019, still 57% below 2009.
- 2020 is excluded because the data ends on 10 Feb 2020.

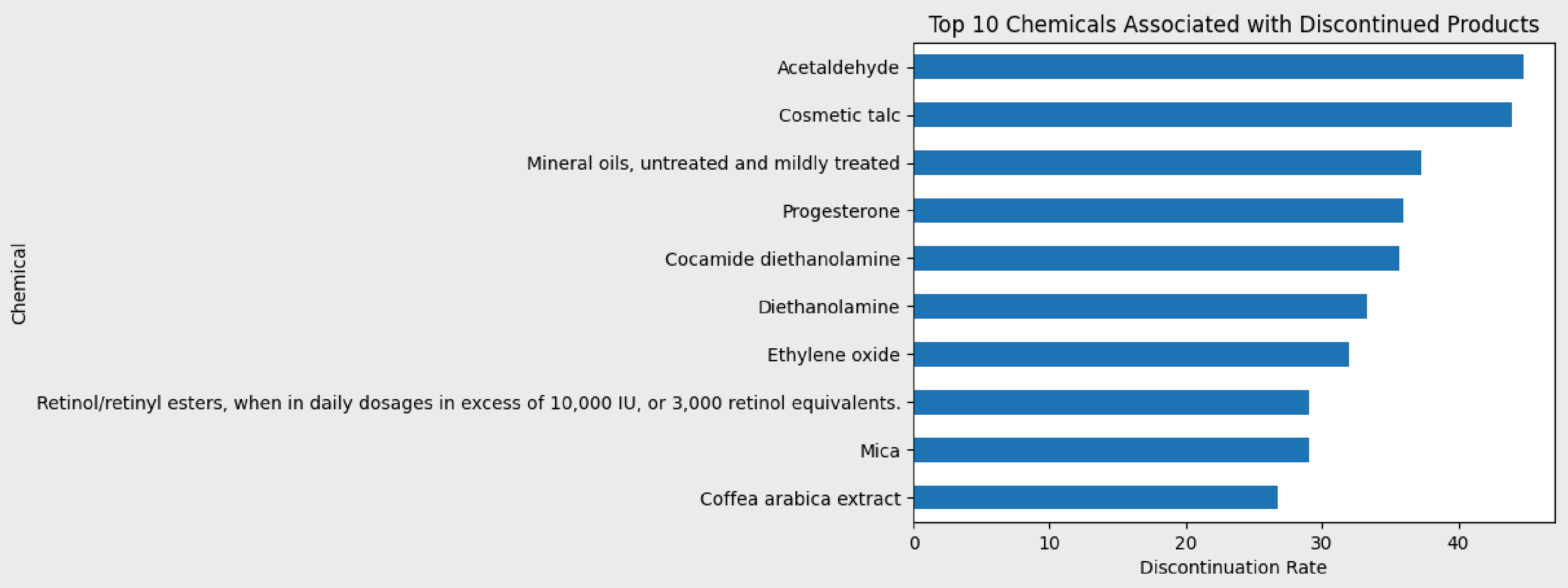
Products Discontinued by Category



Findings

- Personal Care (30%), Baby (24%) and Hair Coloring (22%) are discontinued most.
- Nail (2.4%) and Tattoos (0.4%) are discontinued least.
- Tattoos has the most chemicals per product but the fewest discontinuations.
- Baby has only about 46 products, so its rate is less stable.

Top 10 Chemicals by Product Discontinuation Rate



Findings

- Only chemicals found in 20+ products are ranked.
- Cosmetic talc (116 products), mineral oils (335) and cocamide DEA (898) combine high rates with larger samples.
- Acetaldehyde, progesterone and diethanolamine rest on 25–30 products each.
- This is an association only. The data does not show that any chemical caused discontinuation.

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Products Discontinued by Number of Chemicals



Findings

- The relationship is not a steady rise: 2 chemicals is lowest (9%), 3 (31%) and 4 (53%) are highest.
- Groups with 3 or 4 chemicals hold just 399 of 36.5K products.
- Groups with 5+ chemicals have 15 products in total and were left out.
- Correlation between chemical count and discontinuation is near zero (0.005).

Recommendations

- Provide clearer definitions for fields such as DiscontinuedDate and ChemicalDateRemoved.
- Provide clearer information about duplicate, updated, and removed records.
- Include additional information that could help explain why products are discontinued.

THANK YOU

Thank you for your attention.

